

A Notational Reformulation of Torus-Knot Self-Consistency in Flow-Constraint Variables

Steve Bico Mujjabi, MD*
VuraLabs, Kampala, Uganda
Independent researcher
ORCID: 0009-0001-0556-5516

<https://github.com/bampita-bico/CDFD-Part-I-Release>

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Abstract

This paper records a flow-constraint notation for the torus-knot self-consistency construction used in the earlier CDFD papers. It does not derive a physical vacuum equation of state or establish the physical origin of the twist condition. The relation $\Psi_s = \Phi/C$ is postulated notation whose utility must be judged by an independent physical prediction.

The paper maps selected Fourier quantities onto three CDFD labels: Flow, Constraint, and Adaptive Operating Ratio. The resulting equality is an algebraic restatement of those choices, not an independent result. The paper-local supplementary scripts reproduce the algebra and figures; they do not validate the physical model.

Keywords: Constraint-Driven Field Theory, Constraint-Driven Flux Dynamics, adaptive operating ratio, vacuum structure, Mujjabi tests, reproducible physics, vacuum balance, public CDFD model

1 Project Context

This manuscript constitutes Part I (Paper 10) of the multi-part series *Constraint-Driven Flux Dynamics (CDFD)*. The underlying mathematical architecture builds directly upon the concepts established in Paper 09 of this series [9].

5 Symbol Conventions

CDFD physical-vacuum symbol convention.

*msbico@gmail.com

Symbol	Part I meaning	Physical reading
J or Φ	Driving flux or throughput	Transport current, field throughput, circulation drive, or generic flux; a subscript fixes the exact channel
C	Active constraint or capacity burden	Vacuum transport capacity, geometric confinement, stiffness, boundary resistance, or load-bearing limit
S	Surface responsiveness	Local ability of the vacuum or modeled surface to reconfigure under drive
M_s	Structural memory	Retained geometry, phase history, stiffness history, or accumulated constraint state
Ψ_s	Adaptive operating ratio $(J/C) \cdot S \cdot M_s$ or $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation; it is not a new universal constant
R, a, χ	Ring radius, core radius, and aspect ratio R/a	Geometry used to connect vortex structure to dimensionless electromagnetic scales
ρ_0, c, κ	Vacuum density scale, propagation speed, and transport stiffness	Medium parameters used in stability, pressure, and self-consistency calculations

8 Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Fine-structure constant unless a local equation states otherwise	In the Part I physics papers, un-subscripted α means the electromagnetic fine-structure constant
β, γ	Local response, stiffness, coupling, or correction coefficients	Equation-local coefficients; subscripts bind them to the named mode, channel, or approximation
θ, ϕ, Φ_c	Phase, orientation, or chirality angles	Used for torus-knot orientation, vacuum phase, or chiral phase geometry as defined in the nearby equation
δ, ϵ	Perturbation, residual, tolerance, or small-offset scales	Local stability margins, numerical residuals, or experimental tolerances
λ, μ, ν	Rate, mass, mode, or frequency labels	Meaning is fixed by the equation, subscript, and physics context where the symbol appears
ρ, σ, τ	Density, spread/noise, or time-scale terms	Local density, variance, stress, relaxation, or transport-memory quantities
Ω, Λ	State/domain space or integrated measure where defined	Reserved for explicitly defined state spaces, spectra, or synthesis-level quantities

1 Introduction

Across Papers I–IX [5–13], CDFD introduced a torus-knot parametrisation and several internal algebraic identities. `CORRECTION_STATUS_2026-08-18.md` records the binding distinction between those conditional identities and the unsupported physical mapping.

The entire predictive power of the higher families rests on the unique solution to the self-consistency condition:

$$n\theta_n = \tilde{Q}_n(\theta_n) \equiv \frac{\sum_{k=0}^{n-1} A_k^2}{(\sum_{k=0}^{n-1} |A_k|)^2}. \quad (1)$$

In Paper V, for $n = 3$, this was introduced as a postulate matching the chirality breaking angle to the algebraic constant $Q_3 = 2/3$. In Papers VIII and IX, it was extended to $n \geq 4$ as a working

hypothesis that yielded convergent, mathematically bounded solutions, but lacked a first-principles derivation from the vacuum Equation of State (EOS).

This paper only rewrites the torus-knot construction in the public CDFD flow-constraint notation; it does not close the open physical problem.

2 The Tripartite Vacuum Framework

The CDFD vacuum is modeled as an elastic, transport-capacity-limited continuous medium, in the spirit of Dirac’s physical vacuum [2] and Volovik’s condensate analogy [15]. The vortex dynamics rest on Helmholtz’s conservation theorems [16] and the Faddeev–Niemi knot soliton framework [3]. For this manuscript, the following postulated notational identity is adopted:

$$\Psi_s = \frac{\Phi}{C} \quad (2)$$

where Φ is flow or activity, C is constraint, resistance, or capacity, and Ψ_s is the resulting adaptive operating ratio.

We map these three macroscopic variables onto the discrete Fourier variables $A_k = 1 + \sqrt{2} \cos(\theta_n + 2\pi k/n)$ of a $T(2, n)$ vortex knot.

2.1 Flow (Φ): The Field Energy

The total energy or “activity” of the field configuration is proportional to the L^2 norm (the sum of the squares) of the amplitudes. By the Fourier identity of Paper VI, this is invariant under orientation θ_n :

$$\Phi \propto \sum_{k=0}^{n-1} A_k^2 = 2n. \quad (3)$$

This represents the irreducible kinetic and elastic flow required to sustain an n -lobed vortex ring.

2.2 Constraint (C): The Transport-Capacity Limit

A central postulate of CDFD (Paper I) is that the vacuum possesses a finite transport capacity, which dictates the electromagnetic fine-structure constant $\alpha \equiv e^2/(\hbar c)$. For a multi-lobed knot, the structural stress on the medium is dictated by the total absolute flux. In the discrete basis, the total flux is the L^1 norm $\sum |A_k|$. Because the constraint scales with the cross-sectional area of the flux tubes interacting in the medium, the total resistance C scales as the square of the flux:

$$C \propto \left(\sum_{k=0}^{n-1} |A_k| \right)^2. \quad (4)$$

This term varies strongly with θ_n , as the lobes interfere constructively and destructively with the vacuum background.

2.3 Equilibrium (Ψ_s): The Geometric Twist

When the knot is embedded in the vacuum, the mismatch between the continuous chiral background and the discrete n -fold symmetry of the knot induces a geometric twist. Let θ_n be the phase twist per lobe. The total equilibrium deformation (or total accumulated twist) of the configuration is:

$$\Psi_s = n\theta_n. \quad (5)$$

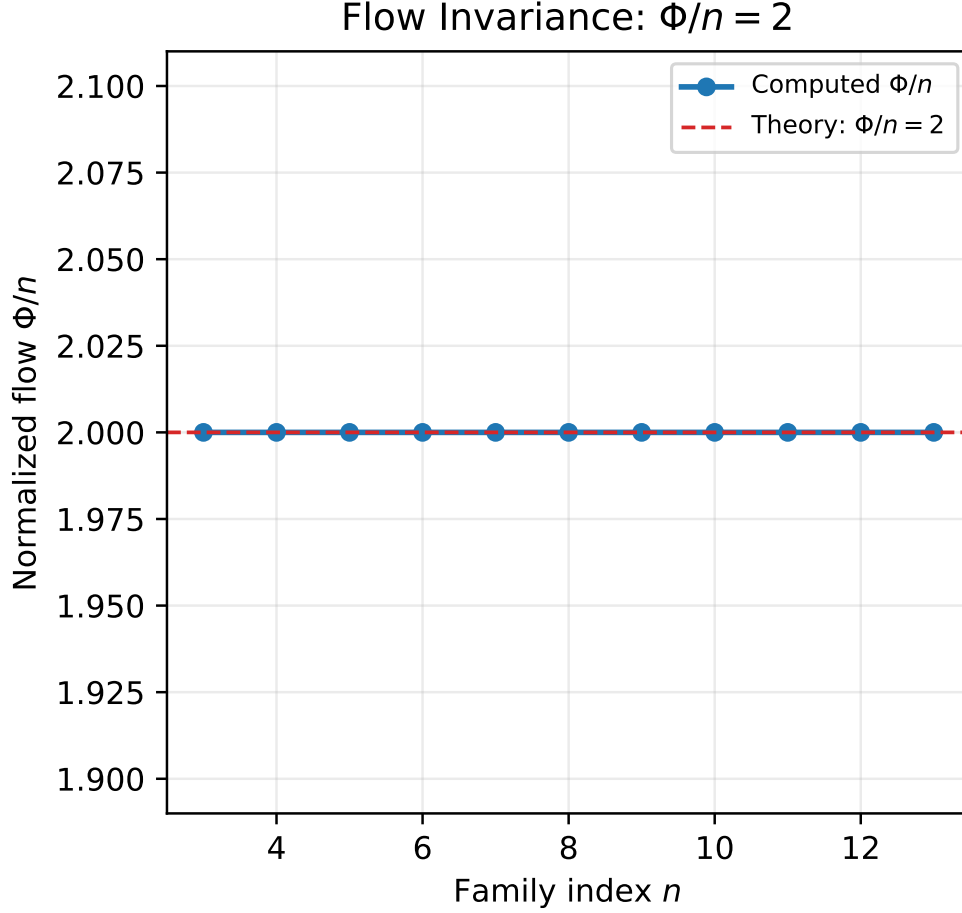


Figure 1: The flow energy ratio Φ/n is exactly 2.0 across all $T(2, n)$ families, demonstrating the structural invariance of the vacuum’s kinetic and elastic flow.

3 The $\Psi_s = \Phi/C$ Notational Identity

Substituting the chosen CDFD mappings (3), (4), and (5) into the postulated notation $\Psi_s = \Phi/C$ gives:

$$n\theta_n = \frac{\sum_{k=0}^{n-1} A_k^2}{\left(\sum_{k=0}^{n-1} |A_k|\right)^2}. \quad (6)$$

The right-hand side is the chosen definition of the modified Koide ratio $\tilde{Q}_n(\theta_n)$ from Eq. (1). This is a consistent relabelling of the construction, not a theorem, proof, or physical derivation.

3.1 Interpretation boundary

The mapping does not show that the observed Koide ratio is a vacuum equation of state, nor that the higher-knot quantities are particle spectra. Any physical interpretation requires an independent mapping and a test that outperforms standard theory.

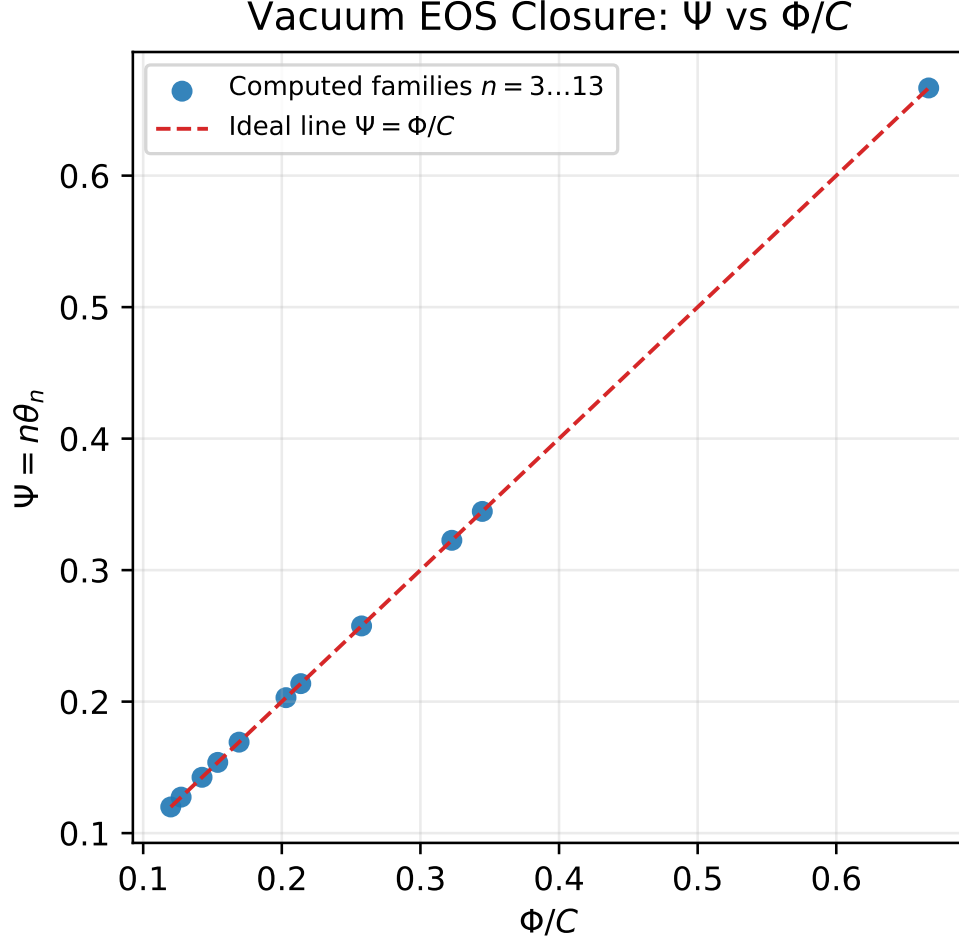


Figure 2: The chosen labels Ψ_s and Φ/C plotted across the stated torus-knot construction. The diagonal identity verifies the internal algebraic relabelling only.

4 Vacuum Engineering as a Hypothetical Control Programme

The public balance law is also the first engineering form of the theory. Vacuum engineering, in this manuscript, does not mean a claimed device. It means a control program: identify which variable in the balance law can be perturbed, then measure whether the response follows the CDFD operator better than the standard model for that domain.

This is why vacuum engineering is introduced here as a future field of study: the disciplined attempt to manipulate, bound, or falsify the state variables Φ , C , S , and M_s in systems where the vacuum or a vacuum-analogue medium is being driven near a transport threshold. The field begins with protocols and null tests, not devices. Its first deliverables are calibrated phase-residual limits, spectral-residual limits, hysteresis bounds, and constraint-threshold maps.

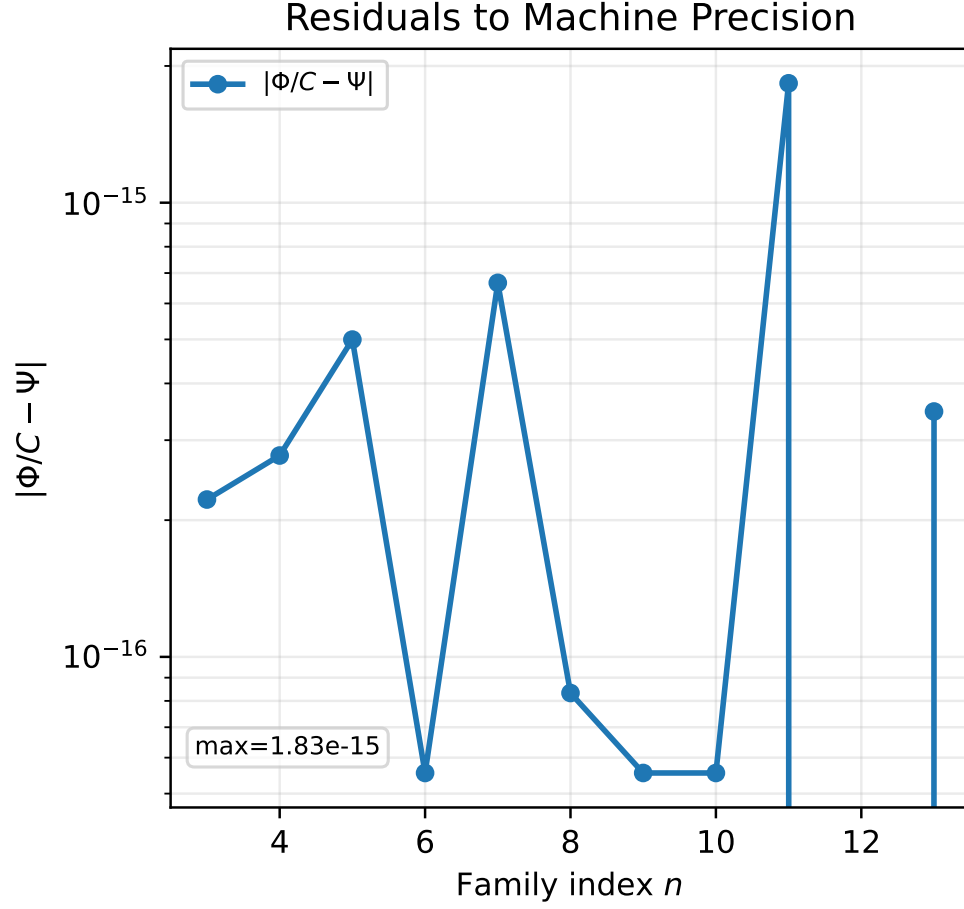


Figure 3: Absolute residuals $|\Phi/C - \Psi_s|$. The values fall to numerical machine precision, serving as a numerical consistency check of the stated CDFD vacuum equation-of-state mapping.

Variable	Engineering handle	Observable	Status
Φ or J	field intensity, current, shear, angular momentum, density gradient	phase, spectral, transport, or force response	controllable in testbeds
C	geometry, boundary, topology, density, confinement	capacity threshold or saturation point	model-dependent
S	susceptibility or response surface	slope, compliance, or relaxation curve	open variable
M_s	prior loading or pulse history	hysteresis, lag, or recovery time τ_M	falsification target

We call this the Mujjabi Vacuum Engineering Principle: steer Φ , C , S , or M_s to move a system toward, across, or away from $\Psi_s = 1$, then measure the residual. Paper XII converts this principle into named laboratory tests.

4.1 CDFD Runtime experiment hooks

The root CDFD Runtime now contains experiment scripts that turn the above principle into machine-readable candidate outputs. These are not empirical confirmations; they are public stress tests that help decide which claims are worth carrying into laboratory protocols.

Runtime output	Observed model-side result	Release use
<code>discovery_vacuum_hysteresis.h5</code>	central M_s peaked at 15.58 after high-flux pulsing	candidate vacuum-memory protocol
<code>discovery_knot_n7.h5</code>	$T(2, 7)$ stability index drifted from 0.992686 to 0.987976	higher-knot stress test
<code>discovery_ool_phase_results.h5</code>	50×50 sweep found four grid points near $\Lambda = 1$	cross-Part-II bridge only
<code>discovery_surface_evolution.h5</code>	W5 ran from 5.15 to 978.99 under overload	instability boundary, not support
<code>frontier_sweep.h5</code>	CODATA 2022 $\chi^* = 137.035999177$ used across 2500 points	hypothesis triage

The publication rule is strict: a runtime file may motivate a paper statement only when the manuscript names the state variables, records the parameters, and states the falsification condition. The refreshed reports in `experiments/reports/` are therefore simulation logs, not evidence that any vacuum-engineering device or clinical intervention already exists.

5 Scope Boundary and Summary

This paper makes no cross-scale biology, medicine, ecology, or Fibonacci claim. Such analogies were removed because formal resemblance does not supply a physical derivation or incremental prediction. The remaining content is a notational restatement of a selected torus-knot construction. It does not close the Part I physical programme, establish a mass hierarchy, or support downstream Parts II and III as a source of established infrastructure.

Supplementary Material

This manuscript is accompanied by release-archive supplementary material in the canonical Part I tree. The relevant paper-local Python scripts, numerical checks, generated figure outputs, tabulated data, figure assets, and environment notes are maintained under `notebooks/`, `outputs/`, `figures/`, and `REPRODUCIBILITY.md` in `Part_I_Fundamental_Physics/`.

Data and Code Availability

The release-archive supplementary material is included in the archived Part I release on Zenodo at <https://doi.org/10.5281/zenodo.20250820>. The current version DOI is assigned by Zenodo after each GitHub release. Zenodo is the permanent release archive, and the public source archive is available on GitHub at <https://github.com/bampita-bico/CDFD-Part-I-Release>. The broader public CDFD Runtime is separate reusable software infrastructure; the numerical values reported here are reproduced from this paper's Supplementary Material.

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Declarations

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Author Contributions

Steve Bico Mujjabi: conceptualization, methodology, formal analysis, software, validation, visualization, writing—original draft, and writing—review and editing.

Conflicts of Interest

The author declares no conflicts of interest.

References

References

- [1] Carl A. Brannen. Spin path integrals and generations. *Foundations of Physics* 40, 1681–1699, 2010. Introduces the parametrisation $m_k = M(1 + \sqrt{2} \cos(\theta + 2\pi k/3))^2$ ($k = 0, 1, 2$) and proves it automatically satisfies the Koide formula.
- [2] Paul A. M. Dirac. Is there an Aether? *Nature*, 168:906–907, 1951. doi: 10.1038/168906a0. Dirac argues the vacuum must be a physical medium with definite velocity at each point; reopens the aether question within quantum field theory.
- [3] Ludwig Faddeev and Antti J. Niemi. Stable knot-like structures in classical field theory. *Nature*, 387:58–61, 1997. doi: 10.1038/387058a0.
- [4] Yoshio Koide. New view of quark and lepton mass hierarchy. *Physical Review D*, 28:252–254, 1983. doi: 10.1103/PhysRevD.28.252. First statement of the charged-lepton mass formula $(m_e + m_\mu + m_\tau)/(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2 = 2/3$; predicted m_τ before precision measurement.
- [5] Steve Bico Mujjabi. Vortex Stability in a Constrained Transport Vacuum and the Origin of the Fine-Structure Constant. Zenodo, 2026. URL <https://doi.org/10.5281/zenodo.21420457>. <https://doi.org/10.5281/zenodo.21420457>.

- [6] Steve Bico Mujjabi. Lepton Masses from Z3-Symmetric Vortex Modes and the Origin of the Koide Formula. Zenodo, 2026. URL <https://doi.org/10.5281/zenodo.21421496>. <https://doi.org/10.5281/zenodo.21421496>.
- [7] Steve Bico Mujjabi. Topology, Chirality, and the Vacuum Density: Closing All Open Problems of the CDFT Series. Zenodo, 2026. URL <https://doi.org/10.5281/zenodo.21421906>. <https://doi.org/10.5281/zenodo.21421906>.
- [8] Steve Bico Mujjabi. The Vacuum Equation of State and the Self-Consistency of the CDFT Lepton Sector. Zenodo, 2026. URL <https://doi.org/10.5281/zenodo.21425048>. <https://doi.org/10.5281/zenodo.21425048>.
- [9] Steve Bico Mujjabi. Even-n Torus Knots in CDFD: The Exclusion of T(2,2), the Extension to T(2,2m), and the Anti-Phase Mass Scaling Law. Zenodo, 2026. URL <https://doi.org/10.5281/zenodo.21435187>. <https://doi.org/10.5281/zenodo.21435187>.
- [10] Steve Bico Mujjabi. Zero Parameters: Deriving the Vacuum Chirality Phase from the Koide Invariant. Zenodo, 2026. URL <https://doi.org/10.5281/zenodo.21427696>. <https://doi.org/10.5281/zenodo.21427696>.
- [11] Steve Bico Mujjabi. The Universal Torus Knot Hierarchy: Zn Symmetry, the Cinquefoil Family, and the General Structure of CDFT Particle Families. Zenodo, 2026. URL <https://doi.org/10.5281/zenodo.21431810>. <https://doi.org/10.5281/zenodo.21431810>.
- [12] Steve Bico Mujjabi. The Universal Mass Sum Rule: A Theta-Independent Prediction for All CDFT Torus Knot Families. Zenodo, 2026. URL <https://doi.org/10.5281/zenodo.21434735>. <https://doi.org/10.5281/zenodo.21434735>.
- [13] Steve Bico Mujjabi. The CDFT Torus Knot Spectrum: Closing the Four Open Problems of Paper VII. Zenodo, 2026. URL <https://doi.org/10.5281/zenodo.21434949>. <https://doi.org/10.5281/zenodo.21434949>.
- [14] William Thomson. On vortex atoms. *Proceedings of the Royal Society of Edinburgh*, 6:94–105, 1869. doi: 10.1017/S0370164600045430. Read April 25, 1867.
- [15] Grigory E. Volovik. *The Universe in a Helium Droplet*. Oxford University Press, 2003. ISBN 978-0-19-850782-5. Vacuum as a quantum condensate; topological defects in superfluid helium as analogues of elementary particles and spacetime.
- [16] Hermann von Helmholtz. Über Integrale der hydrodynamischen Gleichungen, welche den Wirbelbewegungen entsprechen. *Journal für die reine und angewandte Mathematik*, 55:25–55, 1858. doi: 10.1515/crll.1858.55.25. Founding paper of vortex dynamics; proves conservation of vortex tubes in inviscid flow.